19
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Lecture 3 – Data Distributions

INT104 Artificial Intelligence

Data Distributions

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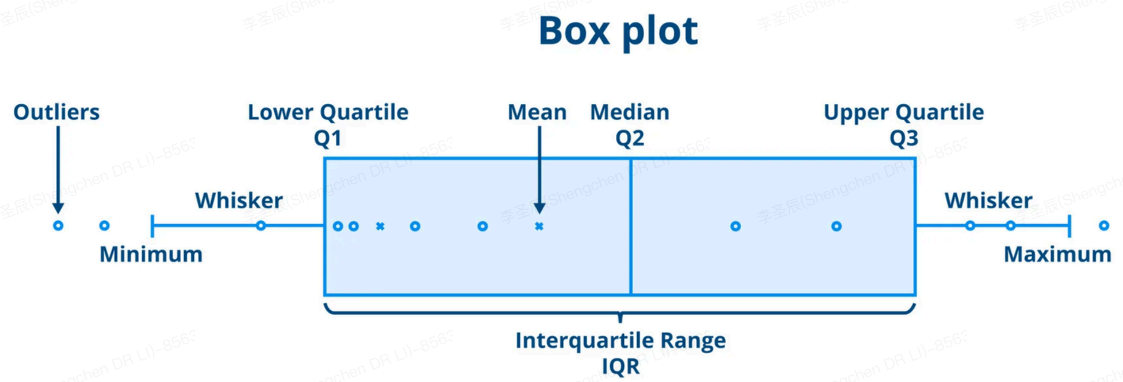
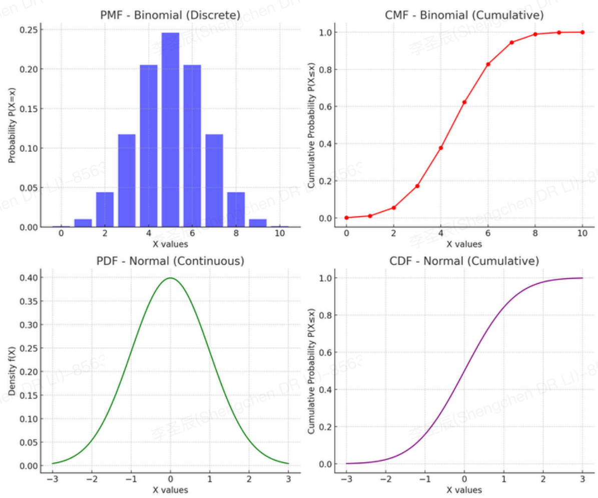
Boxplot (Continued)



Probability Mass Function

- Histogram can be taken as a probability mass function
- Probability mass function can be converted to cumulative mass function for different purposes

> 👨‍🎓 What can a CMF could demonstrate?



Probability

> 📌 What is probability?

- Does low probability lead to impossible consequences?
- How do you understand probability?
  - What is conditional probability?
  - What is joint probability?

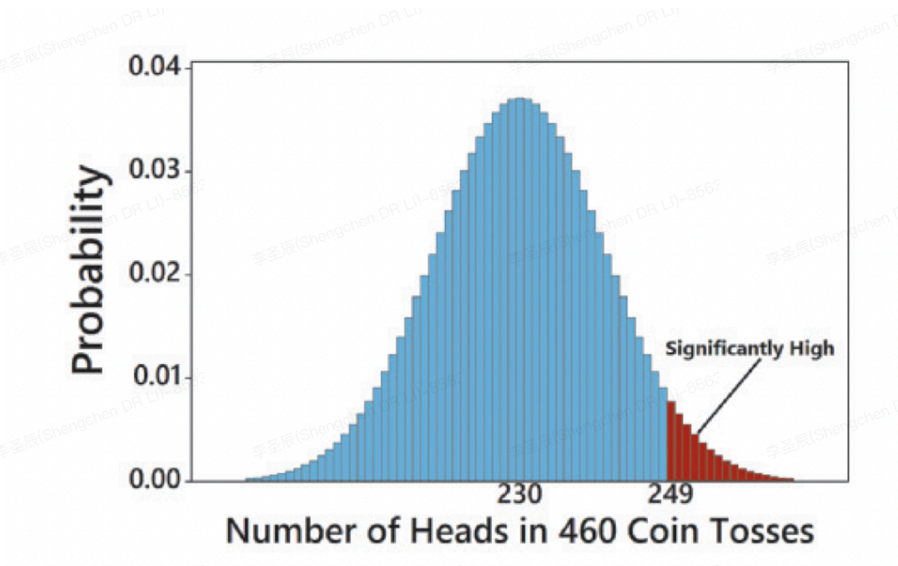
Probability Distribution

- Variable
  - Discrete Variable
  - Continuous Variable
- Probability distribution
- Model

> 📌 How do you measure the certainty of a probability distribution?

Probability Distribution

- Expected Value
- Significant Value



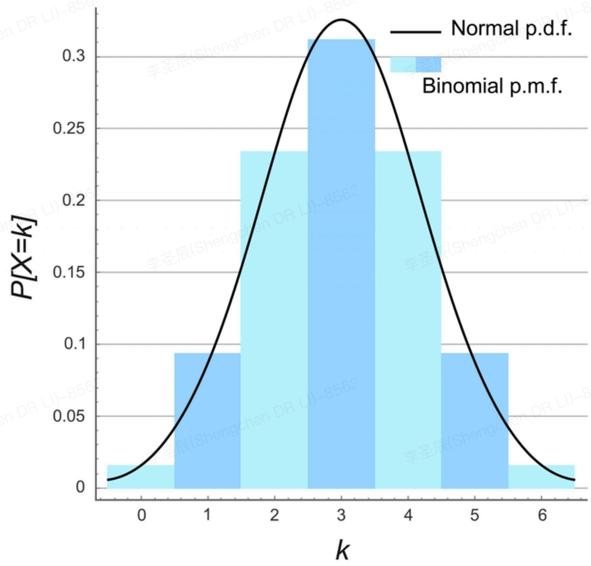
Binomial Probability Distribution

$$P(x) = \frac{n!}{(n-x)!x!} \cdot p^x \cdot q^{n-x}$$

- n → number of trails
- x → number of successes among n trials
- p → probability of success in any one trial
- q → probability of failure in any one trial

🎖️ Binomial Probability Distribution

🏣 Multinomial probability distribution

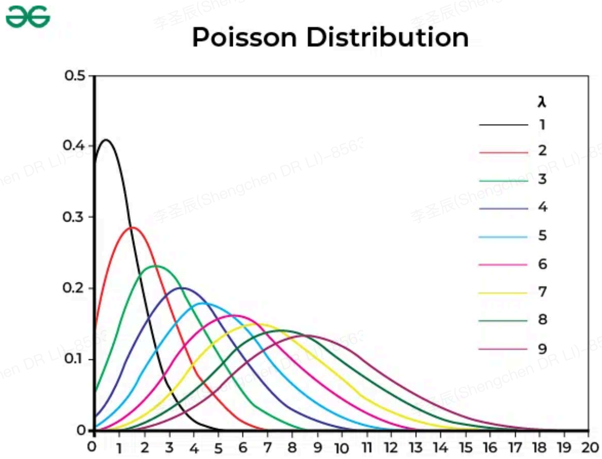


Poisson Probability Distribution

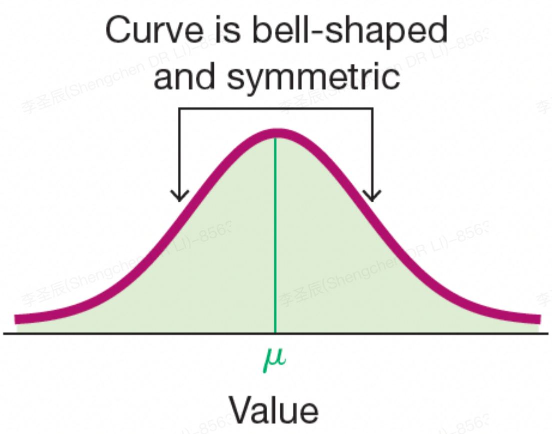
$$P(x) = \frac{\lambda^x \cdot e^{-\lambda}}{x!}$$

λ $\rightarrow$ mean number of occurrences of the event in the intervals

🤯 Poisson Example



Normal Distribution (Gaussian Distribution)



$$y = \frac{e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}}{\sigma\sqrt{2\pi}}$$

👨‍💻 Examples of Normal Distribution

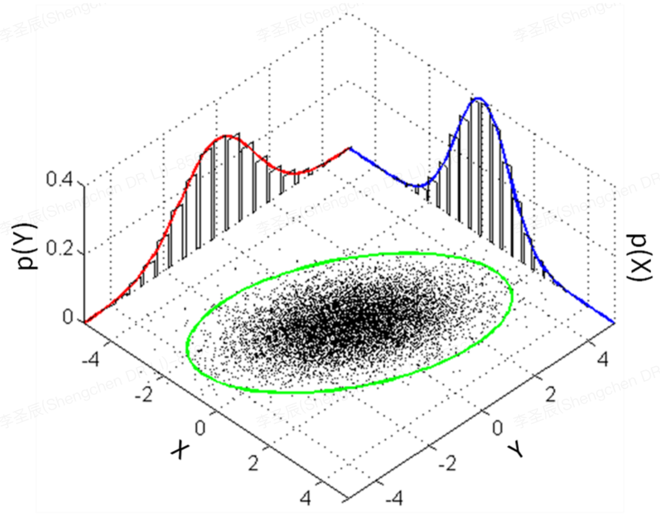
Z Score

📅 Z Score Example

📌 How do you remove the bias of samples to fit normal distribution?

$$x' = \frac{x-\mu}{\sigma}$$

Multi-dimensional Gaussian Distribution



Central Limit Theorem

> 📌 For all samples of the same size n with $n > 30$, the sampling distribution of x can be approximated by a normal distribution with mean μ and the standard deviation $\frac{\sigma}{\sqrt{n}}$.

Key elements:

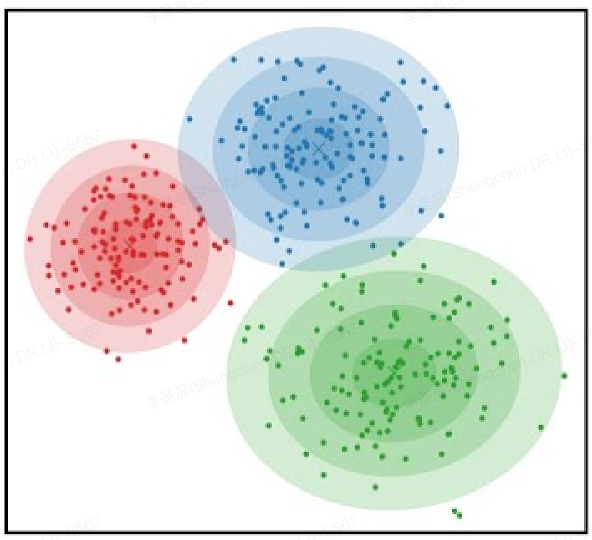
- Population with any distribution has mean μ and standard diviation σ.
- Simple random samples all of the same size n and selected from the population.

> 📌 How to demonstrate variable a having a larger value of variable b

💷 Demonstration of CLT

Gaussian Mixture Model

> 📌 What if the samples are distributed in a clustered way?

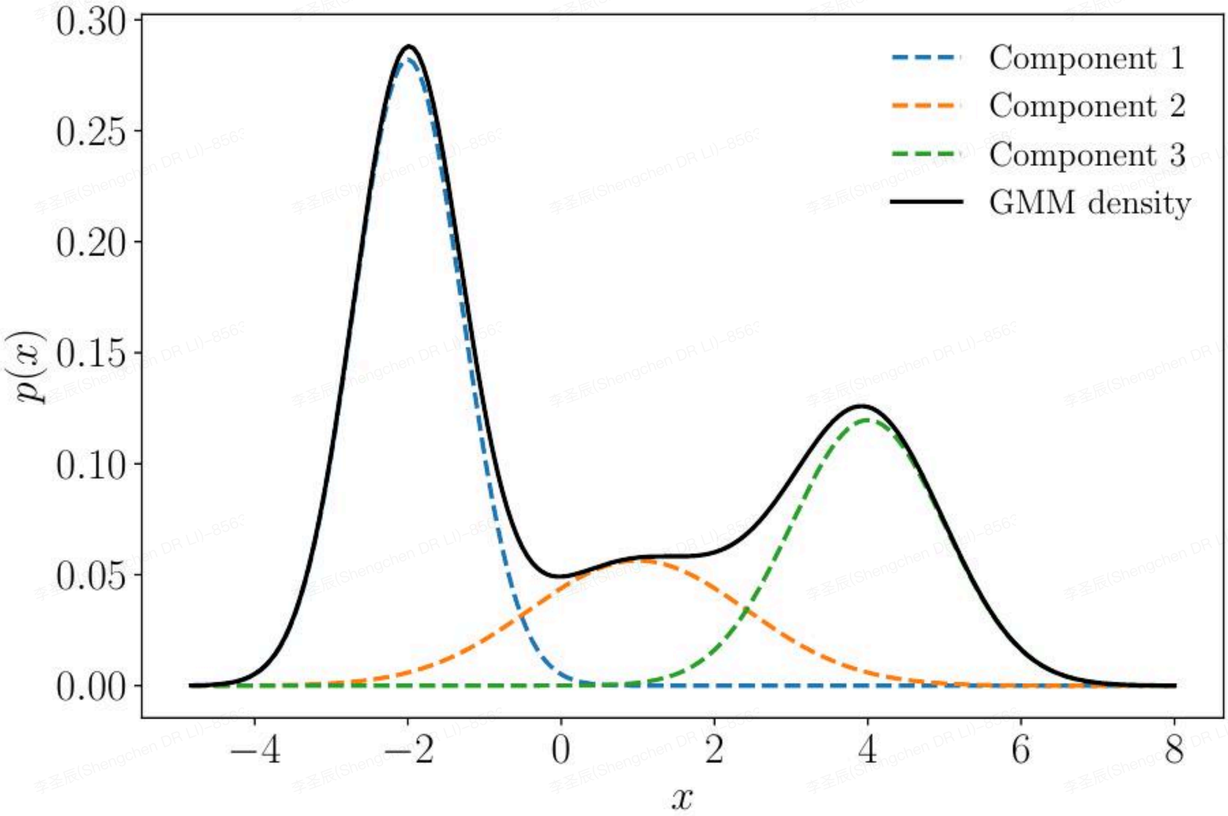


$$p(x) = \sum_{i=1}^{K} \pi_i \mathcal{N}(x|\mu_i, \sigma_i)$$

where

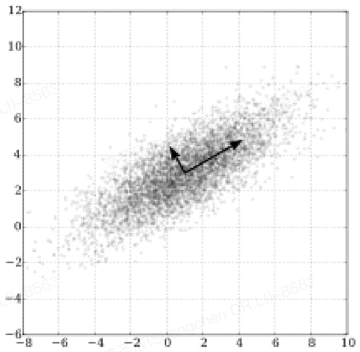
$$\sum_{i=1}^{K} \pi_i = 1$$

GMM

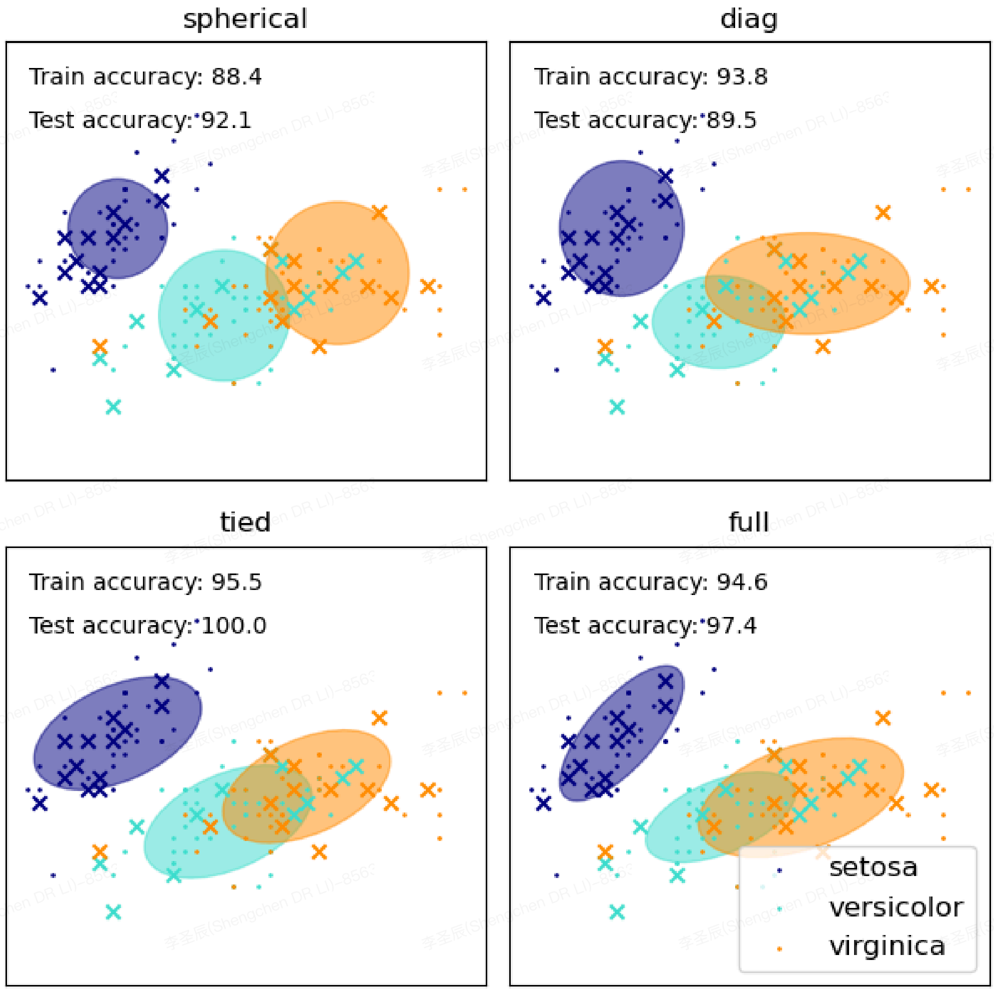


Covariance Matrix

$$
\begin{bmatrix}
Var(x_1) & \cdots & Cov(x_n,x_1) \\
\vdots & \ddots & \vdots \\
Cov(x_n,x_1) & \cdots & Var(x_1)
\end{bmatrix}
$$



Covariance Matrices

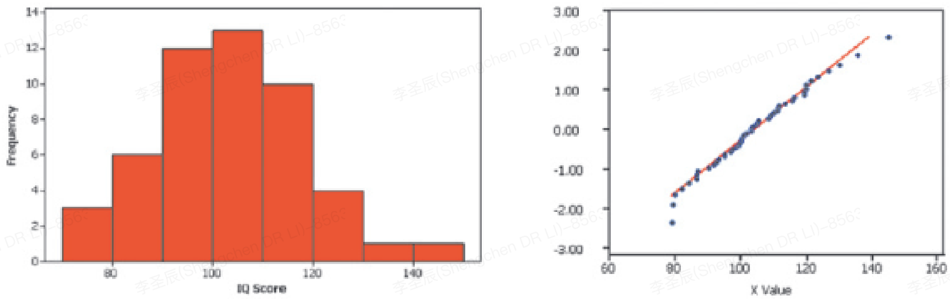


Fit a Model

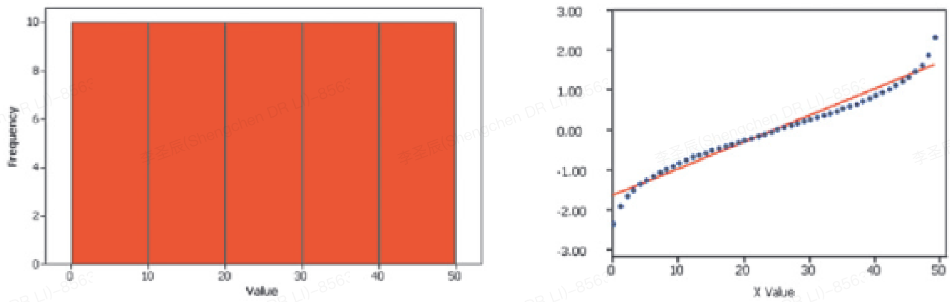
📌 Discussion: how to fit data to a model?

> 📌 Discussion: How to evaluate the fitness of model?

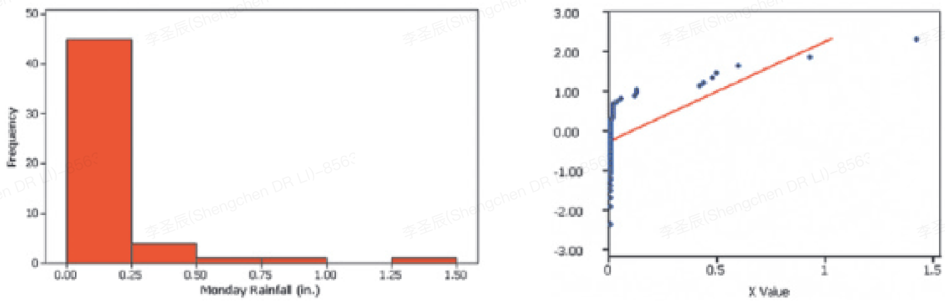
Assessing Normality



Assessing Normality



Assessing Normality



Expectation Maxmisation (EM)

Parameter Estimation:

Typically solved via the **Expectation–Maximization (EM) algorithm**, which iteratively:

2. **M–step**: Updates μ_i , Σ_i , and π_i based on these assignments.

😿 EM Example